Subject: Mathematics | **PERIODIC TEST – 03 (POLYNOMIALS)** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 2: Polynomials** | **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided in Section D.
2. **Section A** comprises 5 questions (4 MCQs and 1 Assertion-Reason) of 1 mark each.
3. **Section B** comprises 3 VSAQ questions of 2 marks each.
4. **Section C** comprises 3 SA questions of 3 marks each.
5. **Section D** comprises 2 Long Answer / HOTS questions of 5 marks each.

SECTION A: Objective & MCQs (5 × 1 = 5 Marks)

Q1. If α, β are the zeroes of $p(x) = 2x^2 - 7x + 3$, then $\alpha^2 + \beta^2$ is equal to: [1]

(A) $\frac{37}{4}$ (B) $\frac{49}{4}$ (C) $\frac{25}{4}$ (D) $\frac{13}{4}$

Q2. A quadratic polynomial with zeroes $5 - 3\sqrt{2}$ and $5 + 3\sqrt{2}$ is: [1]

(A) $x^2 - 10x + 7$ (B) $x^2 + 10x + 7$ (C) $x^2 - 10x - 7$ (D) $x^2 + 10x - 7$

Q3. The number of real zeroes of $f(x) = (x - 2)^2 + 4$ is: [1]

(A) 2 (B) 1 (C) 0 (D) 4

Q4. If α and β are zeroes of $p(x) = 3x^2 - 6x + 4$, then $\left(\frac{\alpha}{\beta} + \frac{\beta}{\alpha}\right) + 2\left(\frac{1}{\alpha} + \frac{1}{\beta}\right) + 3\alpha\beta$ is: [1]

(A) 8 (B) 12 (C) 14 (D) 15

Q5. **Assertion (A):** If the graph of $y = ax^2 + bx + c$ does not intersect the x-axis, then the polynomial has no real zeroes.
Reason (R): Real zeroes correspond to the points of intersection with the x-axis. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Very Short Answer Questions (3 × 2 = 6 Marks)

Q6. Write the family of quadratic polynomials having $-\frac{1}{4}$ and 1 as its zeroes. [2]

Q7. If for a polynomial $p(x)$, the graph of $y = p(x)$ cuts the x-axis at $(-2, 0)$ and $(4, 0)$, write the quadratic polynomial. [2]

Q8. Check whether $g(x) = x - 2$ is a factor of the polynomial $p(x) = x^3 - 3x^2 + 4x - 4$ using the Factor Theorem. [2]

SECTION C: Short Answer Questions (3 × 3 = 9 Marks)

Q9. If α, β are the zeroes of the polynomial $f(x) = x^2 - 3x - 2$, find the value of: [3]

(i) $\alpha^3 + \beta^3$ (ii) $\alpha^2\beta + \alpha\beta^2$

Q10. If one zero of the quadratic polynomial $f(x) = 4x^2 - 8kx - 9$ is negative of the other, find the value of k. Hence find the zeroes of $kx^2 + 3kx + 2$. [3]

Q11. If α, β are the zeroes of $2x^2 - 4x + 5$, find the value of: [3]

(i) $\alpha^2 + \beta^2$ (ii) $(\alpha - \beta)^2$

SECTION D: Long Answer / HOTS Questions (2 × 5 = 10 Marks)

Q12. If the zeroes of the quadratic polynomial $f(x) = x^2 + px + 45$ have a squared difference equal to 144, find the value(s) of p. Also find the zeroes of the polynomial for each value of p. [5]

Q13. Let α and β be the zeroes of the quadratic polynomial $f(x) = x^2 - 3x - 2$. [5]

(i) Find the values of $\alpha^2 + \beta^2$ and $\alpha^3 + \beta^3$.

(ii) Hence, form a quadratic polynomial whose zeroes are $\alpha^2 + \beta$ and $\beta^2 + \alpha$.

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